



VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

25PY101

Engineering Physics

Year 1 Semester 2 Section 24 (ECE + ECEVLSI)

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April 8, 2026

Module Bank

Module 2

Instructions:

1. Answer all questions.
2. BT stands for Blooms Taxonomy.

PART-A

(1) Classical Free Electron Theory (CFET) [10 M]

- (a) A block of Cu is molded into a wire. What is the relationship between the resistance R and diameter d of the wire? [BT 2][1 M]
- (b) Graph the σ vs T for intrinsic semiconductor and metal. [BT 2][2 M]
- (c) If the conductivity of Cu is $5.8 \times 10^7 \text{ S m}^{-1}$ and Au is $4.1 \times 10^7 \text{ S m}^{-1}$ at room temperature, then what is the effective conductivity of wire with core made out of Au and cladding made out of Cu. Here, the cladding diameter is twice of core diameter. [BT 3][2 M]
- (d) If the CFET postulate is assumed that mean free path of electron in a metal is constant, then what is the relationship between resistivity and temperature. [BT 3][2 M]
- (e) The conductivity of Cu and Au at room temperature is $5.8 \times 10^7 \text{ S m}^{-1}$ and Au is $4.1 \times 10^7 \text{ S m}^{-1}$ respectively. The temperature coefficient of resistance of Cu and Au is $3.9 \times 10^{-3} \text{ K}^{-1}$ and $3.4 \times 10^{-3} \text{ K}^{-1}$ respectively. Find the temperature at which the resistivity of Cu equals that of Au. [BT 3][3 M]

(2) Quantum Free Electron Theory (QFET) [10 M]

- (a) Compare and contrast the postulates of classical free electron theory with that of quantum free electron theory. [BT 1][6 M]
- (b) What are the merits of quantum free electron theory? [BT 1][2 M]
- (c) What are the demerits of quantum free electron theory? [BT 1][2 M]

(3) Fermi-Dirac Distribution [10 M]

- (a) Plot the probability of occupancy of energy level in the case of Maxwell-Boltzmann distribution and Fermi-Dirac distribution at [BT 2][2 M]
 - (i) $T = 0 \text{ K}$
 - (ii) $T \neq 0 \text{ K}$
- (b) Five free electrons exist in a three-dimensional infinite potential well with all three widths equal to $a = 12 \text{ \AA}$. [BT 3][4 M]
 - (i) Determine the Fermi energy level at $T = 0 \text{ K}$.
 - (ii) Repeat part (i) for 13 electrons.
- (c) Fermi level of a metal is 6.25 eV and electrons in this material follow the Fermi-Dirac distribution. Calculate the temperature at which there is 1% probability that a state 0.30 eV below the Fermi energy level will not contain an electron. [BT 3][4 M]

(4) Electronic specific heat of solids [10 M]

- (a) Define specific heat of a material. What is its SI unit? [BT 1][2 M]

- (b) Specific heat of mercury $^{200}_{80}\text{Hg}$ is $0.14 \text{ J g}^{-1} \text{ K}^{-1}$. Report its specific heat in SI units. [BT 1][2 M]
- (c) Derive the expression for specific heat capacity from classical free electron theory and quantum free electron theory. [BT 4][6 M]
- (5) E vs k diagram [10 M]
- (a) Sketch the E vs k diagram for a free electron. [BT 1][1 M]
- (b) Sketch the E vs k diagram for an electron in a periodic potential. [BT 1][1 M]
- (c) Consider an intrinsic semiconductor with all electrons occupying states in the lowest valence band at $T = 0 \text{ K}$. Plot the E vs k showing occupancy of levels at both $T = 0 \text{ K}$ and $T \neq 0 \text{ K}$. [BT 2][2 M]
- (d) Two possible conduction bands are shown in the E vs k diagram given in Figure. 1. State which band will result in the heavier electron effective mass; state why [BT 2][2 M]

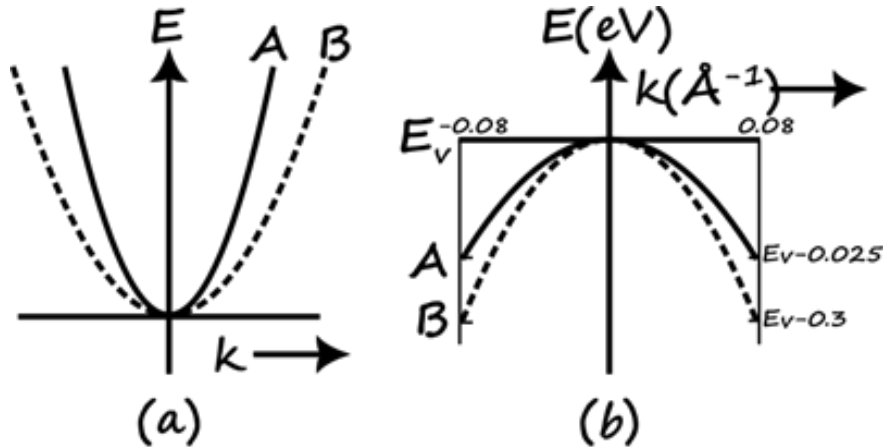


Figure 1: (a) Conduction and (b) valence bands of materials A and B.

Figure 1:

- (6) Fermi level in semiconductors [10 M]
- (a) Sketch the energy band diagram showing the density of states, Fermi-Dirac probability function and areas representing electron and hole concentration for an intrinsic semiconductor at $T \neq 0 \text{ K}$. Assume electron effective mass is equal to hole effective mass. [BT 1, 2][3 M]
- (b) Calculate position of Fermi level with respect to center of the bandgap in silicon for [BT 2][2 M]
- (i) $T = 0 \text{ K}$
- (ii) $T = 300 \text{ K}$
- The electron effective mass is $1.08m_0$ and hole effective mass is $0.56m_0$.

- (c) Draw, compare, and contrast the sketches of the energy band diagram for p-type semiconductor at $T = 0\text{ K}$ and $T = 300\text{ K}$. [BT 3][2 M]
- (d) Silicon atoms, at a concentration of $7 \times 10^{15}\text{ cm}^{-3}$, are added to GaAs. Assume that the silicon atoms act as fully ionized dopant atoms and that 5% of the dopants replace Ga atoms and 95% replace As atoms. [BT 2][3 M]
- Determine the donor and acceptor concentrations.
 - Is the material n type or p type?
- (7) p-n junction diode [10 M]
- Write the equation and sketch the $I - V$ characteristics of pn junction diode.[BT 1][2 M]
 - Define reverse saturation current of pn junction diode. Derive it from the $I - V$ characteristics. [BT 1][2 M]
 - A silicon diode can be used to measure temperature by operating the diode at a fixed forward-bias current. The forward bias voltage is then a function of temperature. Assume that the reverse saturation current is proportional to $\exp(-E_g/k_B T)$. At $T = 300\text{ K}$, the diode voltage is found to be 0.60 V . Determine the diode voltage at $T = 310\text{ K}$. What is the percentage change in voltage? [BT 4][3 M]
 - The pn junction diode can also be used as a temperature sensor by operating the diode at a fixed reverse-bias voltage. What is the percentage change in current for the same temperature rise? [BT 4][2 M]
 - Which mode is more sensitive, the constant forward current mode or constant reverse voltage mode? [BT 2][1 M]
- (8) Solar cell [10 M]
- Sketch the $I - V$ characteristics of pn junction solar cell. [BT 1][1 M]
 - Define short circuit current and open circuit voltage. [BT 1][2 M]
 - A silicon solar cell under standard test conditions has short circuit current density of 35 mA cm^{-2} , and open circuit voltage of 0.60 V . The fill factor for the material is 0.80 . If the incident power is 100 mW cm^{-2} , calculate the efficiency of the cell. [BT 3][4 M]
 - The energy band gap of a semiconductor is 3 eV . Is this material useful for solar cell applications? Give reasons. [BT 4][3 M]
- (9) Direct and indirect band gap semiconductors [10 M]
- Sketch the E vs k diagram for [BT 1][2 M]
 - a direct bandgap semiconductor, and
 - an indirect bandgap semiconductor.

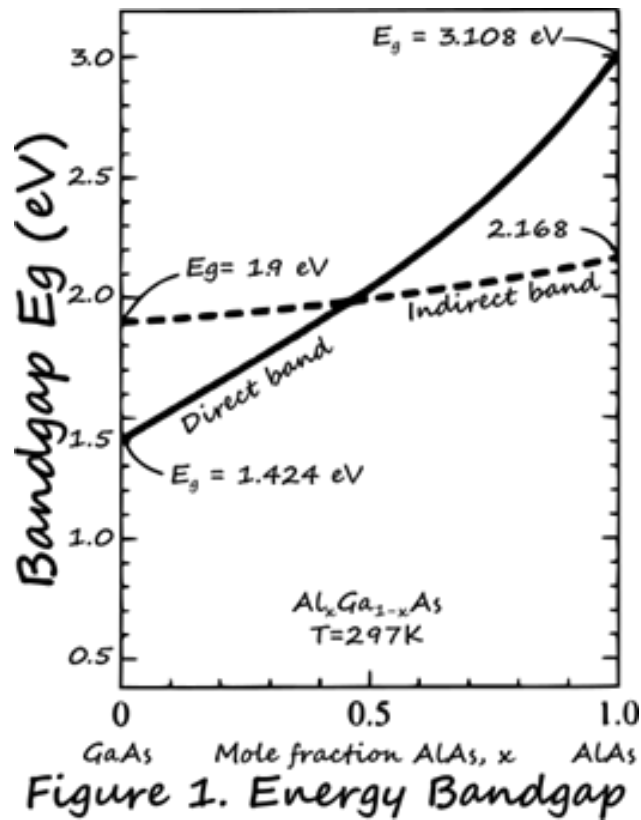


Figure 2:

- (b) Why is direct band gap semiconductor useful for solar cell application? Crystalline silicon is indirect band gap semiconductor. What is the process employed in industry to “convert” it into a direct band gap semiconductor for solar cell application? [BT 2][2 M]
- (c) c) Which material is used in blue LED? What is the bandgap of the material? [BT 3][3 M]
- (d) $Al_xGa_{1-x}As$ is a ternary semiconductor and the energy band gap variation as a function of mole fraction x of AlAs is shown in Figure. 2. Which composition can be used as material for blue LED application? [BT 4][3 M]

(10) LED [10 M]

- (a) What are the materials used for red, green, and blue LEDs? [BT 1][2 M]
- (b) What are the mechanisms that decrease the efficiency of an LED? [BT 1][3 M]
- (c) The energy band gap variation as a function of mole fraction x of $GaAs_{1-x}P_x$ is shown in Figure. 3. What is the range of colours that can be emitted if used for LED application? Report in wavelength. [BT 3][3 M]
- (d) Calculate the critical angle at GaAs - air interface. The refractive index of GaAs is 3.8 at $\lambda = 0.70 \mu m$. [BT 3][2 M]

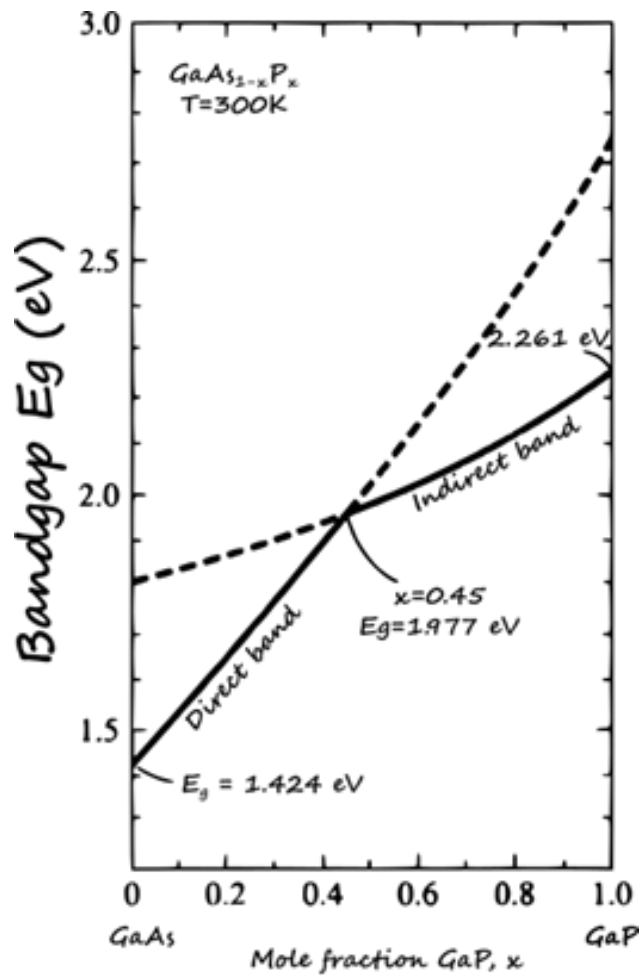


Figure 1. Energy band gap

Figure 3: